

Assignment 2 – CS60016 AI Ethics

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Dataset and Features

Bank Marketing dataset (45,211 instances, 16 features)

- Target y : yes/no $\rightarrow$ 1/0
- Sensitive attribute z : age $\geq$ 39 $\rightarrow$ 1, else 0
- Train/Test split: 80/20 (stratified)
- Embedding dimension: 49

Interpretable features: age_bin, balance_bin, housing_bin, education_bin, marital_bin, loan_bin

Task X: Model Training

Model	Test Accuracy	p%-rule
Model-Accurate	90.42%	91.73%
Model-Fair	90.42%	92.35%

Observation: Both models achieve identical accuracy, but Model-Fair improves the p%-rule, indicating reduced disparity across sensitive groups.

Instance Selection

Selected 5 instances:

- DISAGREE: 149, 405
- AGREE: 0, 1, 2

Task Y: Perturbation

- $N = 10$ perturbed samples per instance (as specified)
- Random bit-flipping on interpretable features
- Non-interpretable features kept fixed

Justification: Binary flipping preserves interpretability and ensures local neighborhood exploration consistent with LIME.

Task Z: LIME

- Weight: $w(z) = \exp(-D(x', z)^2/\sigma^2)$
- $\sigma^2 = 4.5$ (proportional to feature dimension)
- Linear model fit using weighted least squares

Result: Weighted $R^2 \approx 0.99 - 1.00$ for all instances, indicating strong local fidelity.

Local Fairness Analysis

Instance	Age Importance (Accurate)	Age Importance (Fair)
1	-0.0544	-0.0499
2	-0.0466	-0.0410
3	-0.0564	-0.0526
4	-0.0234	-0.0211
5	-0.0393	-0.0348

Observation: Across all instances, the magnitude of age_bin importance consistently decreases in Model-Fair. This confirms that the fairness constraint reduces local reliance on the sensitive attribute.

Submodular Optimization

Model	Top-1	Top-3 Features
Model-Accurate	education_bin	education, housing, balance
Model-Fair	education_bin	education, housing, loan

Observation: The sensitive feature (age_bin) is not selected in top-3 features for either model, further indicating reduced global importance.

Discussion of Results

- The fairness constraint improves p%-rule without affecting accuracy, indicating that the model redistributes feature importance rather than sacrificing performance.
- LIME explanations show a consistent reduction in dependence on the sensitive attribute across all local instances.
- High R^2 values confirm that local linear models accurately approximate the black-box predictions.
- Submodular selection highlights education_bin and housing_bin as dominant features, aligning with LIME results and ensuring global consistency.
- These findings align with the intuition from Ribeiro et al. (2016), where locally faithful explanations reveal feature-level behavior not visible from global metrics.

Key Insight

The Model-Fair achieves fairness not by removing predictive power, but by reducing reliance on the sensitive attribute both locally (LIME) and globally (submodular selection).